TRIPLES, FACTOR PAIRS AND A CONJECTURE

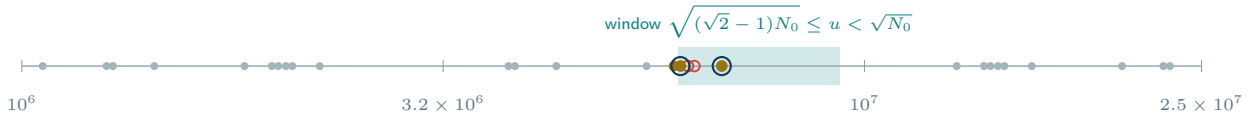
At most two triangles per annulus

Exact constraints, a calibrated model and a search to $c \le 10^8$ suggest no annulus has three primitive triangles.

1 / THE EXACT REDUCTION

Each triangle with annulus value N is an allowed factor pair of N that passes a square test

For a primitive triangle $N = \frac{4A}{\pi} = uv$ with $u = c - 2r$ and $v = c + 2r$, coprime and odd. A factor pair gives a triangle exactly when (i) $\sqrt{(\sqrt{2}-1)N} \le u < \sqrt{N}$, the window from $1 < v/u \le 1+\sqrt{2}$; (ii) the prime factors $\equiv 3 \pmod 8$ lie in u, those $\equiv 7$ in v, none is $\equiv 5$, and those $\equiv 1$ may go either way; (iii) $u \equiv v \pmod 4$; and (iv) $D = \frac{1}{2}(u^2 + 2uv - v^2)$ is a perfect square, when the legs are $\frac{v \mp \sqrt{D}}{2}$. On the curve $E_N : Y^2 = X^3 + 4NX^2 - 4N^2X$ with $X = 2u^2$, the window is the arc from the point of order two at $X = 2(\sqrt{2}-1)N$ to $X = 2N$.



divisors of $N_0 = 87\,110\,147\,844\,489$ near the window, log scale: grey break the mod-8 rule, red rings sit in the window but break it, gold are allowed, and the two ringed in navy pass the square test

2 / A CHANCE MODEL FOR THE SQUARE TEST

Treat each allowed pair as a biased coin, and check the bias against the real count

A random integer near D is a square with chance about $\frac{1}{2\sqrt{D}}$. The square test is less random than that: $D \equiv 1 \pmod 8$ always, which multiplies the chance by 4, and D is automatically a quadratic residue modulo every prime of N, which multiplies it by 2 for each of the $\omega(N)$ primes. So a candidate passes with probability

$$p \approx \kappa \, \frac{2^{\omega(N)+1}}{\sqrt{D}}$$

with one fitted constant, $\kappa = 0.908$.

range of N	model, $\kappa = 1$	actual triangles	ratio
10^5 to 10^6	126.7	111	0.876
10^6 to 10^7	399.0	368	0.922
10^7 to 10^8	1259.6	1140	0.905
10^8 to 10^9	3978.9	3607	0.907
10^9 to 10^{10}	12 601.9	11 443	0.908

Summed over every allowed pair with $N \le 10^{10}$, the model with $\kappa = 1$ over-counts the true triangles by a steady 9% in every decade, so κ is the only adjustment.

3 / WHAT THE MODEL PREDICTS

Shared values keep appearing slowly, and triples almost never

range of N	expected shared values	expected triples	found
up to 10^{10}	0.62	2×10^{-5}	0
10^{10} to 10^{12}	0.56	2×10^{-5}	0
10^{12} to 10^{14}	0.85	4×10^{-6}	1
10^{14} to 8.28×10^{15}	1.12	6×10^{-7}	1
total, searched	**3.2**	4×10^{-5}	**2**
beyond, per decade	about 0.9	10^{-7} and falling	not searched

The expected numbers are $\sum_N \sum_{i<j} p_i p_j$ for shared values and $\sum_N \sum_{i<j<k} p_i p_j p_k$ for triples, computed exactly up to $N = 10^{10}$ and beyond that from 40 000 to 80 000 sampled triangles per decade. The model expects about three shared values where the search found two, and about 5×10^{-7} triples beyond the searched range, uncertain by a factor of three.

4 / WHY PLAUSIBILITY FALLS WITH SIZE

A partner is rare, and a second partner rarer still

Each candidate passes with chance of order $2^{\omega(N)}/\sqrt{N}$, while a decade near N holds about $\sqrt{N}$ triangles. A shared value needs one further pass, so its expected count per decade stays roughly constant, creeping upwards: infinitely many are expected, growing like a power of $\log N$. A triple needs two further passes, so its expected count per decade carries an extra factor of about $N^{-1/2}$ and the total over all N converges, most of it lying below the searched range, where none exists.

5 / THE CONJECTURE

What is proved, what is computed, and what is conjectured

Proved. A triangle with annulus value N has $c \le \sqrt{N/(2\sqrt{2}-2)}$, so the exhaustive search over all 15 915 492 primitive triangles with $c \le 10^8$ settles every $N \le 8.28 \times 10^{15}$: no such N has three primitive triangles and exactly two have two. For N_0 there are exactly two at any size.

Conjecture. (a) No positive integer is the annulus value $4A/\pi$ of three distinct primitive Pythagorean triangles. (b) Infinitely many are the annulus value of exactly two, and the number of these up to X grows more slowly than any power of X.

The model assumes the square tests behave independently; a polynomial family of triangles, or a structured supply of points on the arcs of the curves E_N, would break that assumption and is the most likely way the conjecture could fail. A search to $c \le 10^9$ should find about two more shared values and no triple.

ASIDE / A TEST YOU CAN RUN

List the allowed pairs in the window of any shared value and test each: only two pass, as for N_0.

LINKS TO OTHER SHEETS

The annulus sheet defines N; the two-triangles sheet proves the case of N_0; the Pell sieve sheet explains the mod-8 side rule.

Convention. $N = 4A/\pi = uv$; $\omega(N)$ the number of distinct primes of N; κ fitted on $10^8 \le N \le 10^{10}$; expected counts are model values, the found column is exact.

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